

ADVANCED COMPUTER NETWORKS
(MTITE101)

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Practical 3:

Aim: To construct a network topology in **ns-3** using Python bindings to demonstrate **IPv6 Stateless Address Autoconfiguration (SLAAC)**, **Neighbor Discovery Protocol (NDP)**, and end-to-end packet routing between network nodes.

2. Step-by-Step Implementation Guide

Step 1: Environment & Directory Setup
Open a terminal and navigate to the root directory of your ns-3 installation:

Bash
cd ~/ns-allinone-3.33/ns-3.33

The scratch/ directory in ns-3 is reserved for user scripts. Any script placed here is automatically detected and built by the Waf build system.

Step 2: Script Creation
Create a new file named ipv6-slaac.py inside the scratch/ directory

Step 3: Script Execution
Execute the Python script through the ns-3 Waf tool:

Bash
./waf --pyrun scratch/ipv6-slaac.py

3. SLAAC & Neighbor Discovery Protocol (NDP):

Behind the scenes (visible when log level is increased or pcap is captured):

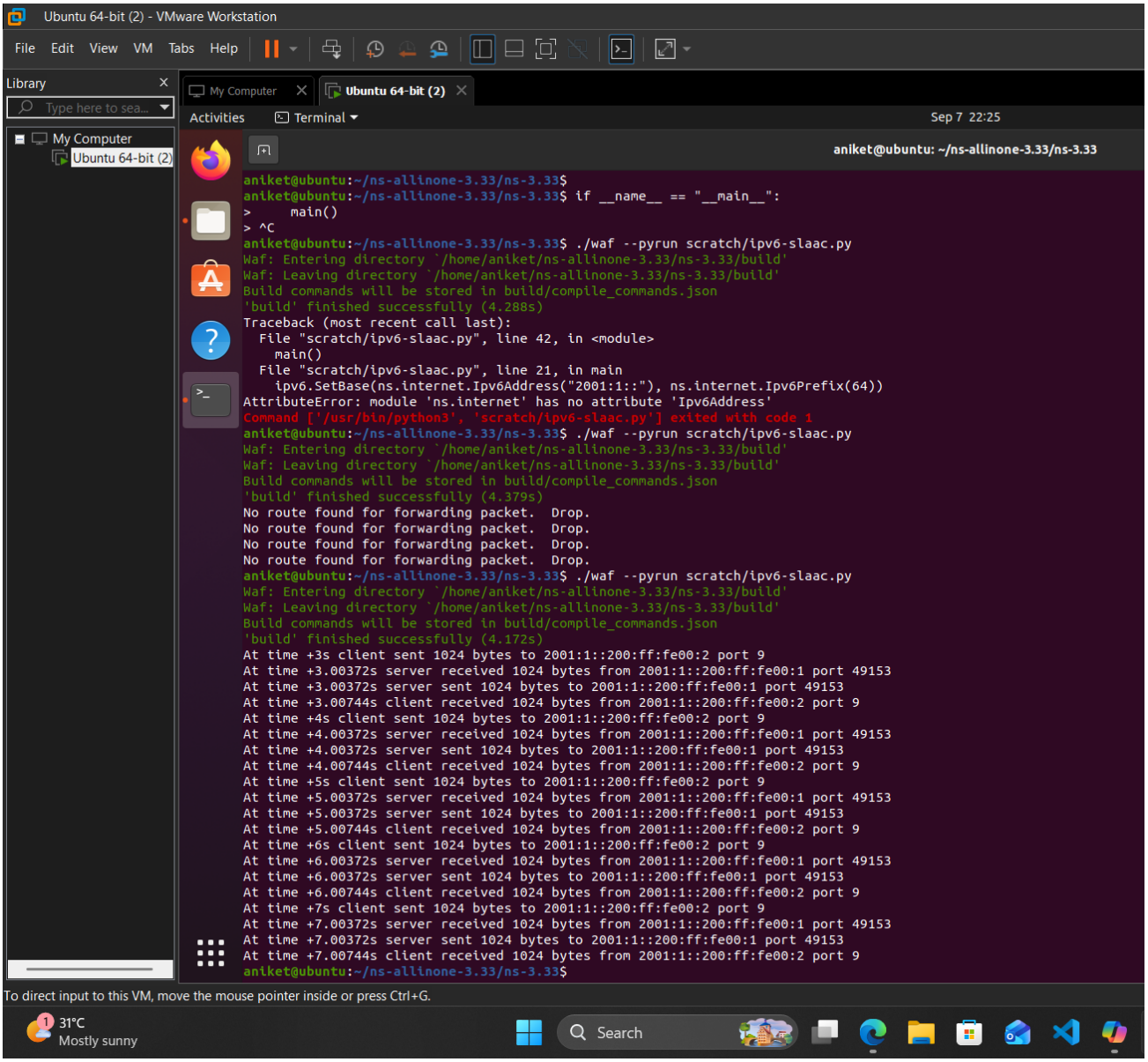
Link-Local Address Generation: Each node automatically configures a link-local address starting with fe80:: based on its interface MAC address (EUI-64 standard).

Router Solicitations (RS) & Router Advertisements (RA):

Node 0 acts as the IPv6 router (interfaces.SetForwarding(0, True)).

Node 0 broadcasts a Router Advertisement (RA) containing the network prefix 2001:1::/64.

Node 1 receives the RA and autoconfigures its global unicast IPv6 address (2001:1::200:ff:fe00:2).



Explanation:

Neighbor Solicitation / Advertisement (NS/NA): Prior to sending the first payload at 3.0s, Node 0 performs Neighbor Discovery (the IPv6 equivalent of ARP) to map Node 1's IPv6 address to its Layer 2 MAC address.

Data Transfer:

Time 3.0s: Node 0 sends a 1024-byte UDP Echo Request.

Time 3.00257s: Node 1 receives the packet after a 2ms channel delay plus transmission delay, then sends an Echo Reply.

Time 3.00514s: Node 0 receives the response, confirming active two-way reachability via SLAAC.